Clinical EEG-to-fMRI feasibility
PhD thesis exploration

Exploratory research | reviewed M3 experiments | 21 October 2026

The clinical question

Which EEG-to-BOLD estimates could support one clinical action? Test regional fidelity and failure modes before considering patient-specific fMRI replacement.

A computer science thesis to explore

Thesis aim: identify which BOLD information is recoverable, then build models that generalise and know when to abstain. This explores a possible computer science PhD.

Candidate contributions: multimodal system identification; robust person/site transfer; calibrated, selective regional prediction. Establish novelty against UnEBOLT, MARBLE and UniEFS, rather than promising clinical replacement.

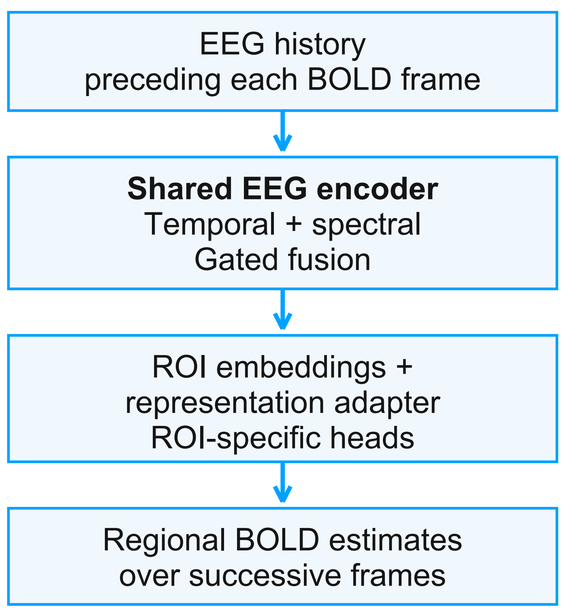
First application: a wakefulness advisory prompting technologist review. Compare direct arousal prediction: BOLD synthesis must earn its complexity. One regional feedback target is a conditional second route. Diagnostic mapping needs a separate validation programme.

Working hypothesis

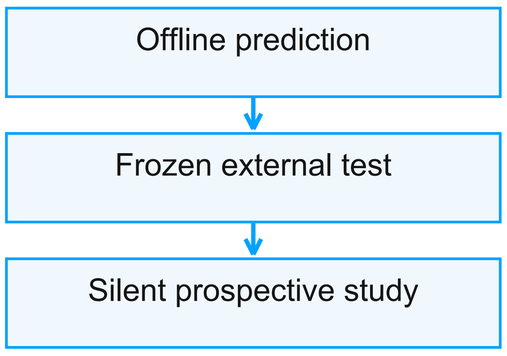
Predictive value may depend more on measured information and target definition than on model size. Cross people, duration, sensors and capacity on fixed holdouts. A nonlinear gain would favour richer models; an added-sensor gain would favour acquisition. Null results and small learning curves cannot establish a biological ceiling.

Model architecture

UnEBOLT schematic [1]



Published reference; not tested here.
Patient BOLD/utility unvalidated.



Proposed validation pathway

Models and possible uses

Healthy technical BOLD evidence; uses proposed

UnEBOLT [1] | 64 regions, r .418.
Proposed: one regional feedback target.

UniEFS [2] | 512 regions, r .367.
EEG-only Parkinson classification probe;
patient BOLD unvalidated.

MARBLE [3] | 64 regions, r .434; 2.6M parameters.
Proposed: compact regional prediction.

Reported temporal r; different protocols, not a ranking.

M3 evidence and proposed PhD next steps

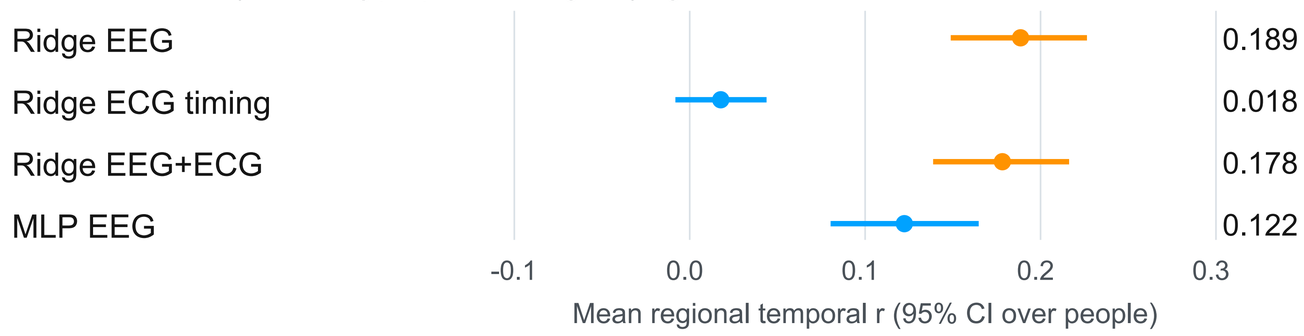
M3 study design

ds006040 development: 27 people/54 runs; three outer folds, 96 fitted outer cells. Inner tuning separate. Matched inputs: 300 EEG-power features, ECG timing/availability or both; ridge/small MLP. Cross 4/12 training people x 120/240 retained target-seconds. One seed; four equal-weight MLP target groups. Published models were not reproduced.

Ridge EEG regional gain over its predicted-global template is uncertain: paired delta-r .004, 95% CI -.008 to .017.

New M3 empirical evidence

EC-trained/test-EC | 12 training people x 240 target-s | regional64



27 people; person-bootstrap 95% CIs. EEG ridge predictive R-squared .028 in normalized targets.

People and retained target time

ds006040 | EC-trained/test-EC | n=27 | regional64 temporal r

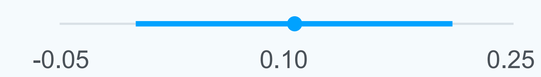


Person-bootstrap 95% CIs; fixed subsets and EC evaluation.

NATVIEW: transfer and QC

Strict timing 1/8; amended QC 6/8.
Secondary exploratory; n=6; pilot excluded.

Regional r 0.104
95% CI [-0.0002, 0.207]



R2 -0.071; 740/2304 frames (32.1%).
Frozen VU model/thresholds; 23 target sites include interpolation.

Three proposed work packages

WP1 | Measurement design. Rank montage, history and sensors; test held-out gains against cost-matched choices.

WP2 | Structured estimation. Learn shared dynamics with counted calibration. Test regional gains beyond ridge and state/physiology controls.

WP3 | Selective prediction. Report error at useful coverage under stated shift assumptions; compare quality rejection.

Clinical uses and limits

Clinical precedent: Prism's fMRI-informed EEG biofeedback is cleared for adjunctive PTSD treatment; diagnostic interpretation is excluded [4].
Proposed: wakefulness support or one regional feedback target. Disease classification does not verify patient BOLD recovery.
M3 limits: inherited preprocessing is retrospective; no absolute %BOLD calibration. Causal inputs do not establish end-to-end live use. Sparse ECG; one development site/seed; clinical benefit untested [5].

[1] UnEBOLT, MIDL 2026. **[2]** UniEFS, ICML 2026.
[3] MARBLE, MICCAI 2026. **[4]** FDA K222101, 2023.
[5] DECIDE-AI, 2022. Full sources: accompanying report.

Seamus Taylor | PhD thesis exploration